Tankala Sai Prasanth

Software Engineer | AI & Agentic Systems Developer | Data & Cloud Solutions

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PROFESSIONAL PROFILE & CORE PHILOSOPHY

Multi-disciplinary Software Engineer and AI Practitioner with expertise spanning Autonomous Agent Orchestration, Hierarchical Multi-Agent Systems (HMAS), Enterprise Process Modernization (Power Platform & Copilot Studio), and Full-Stack Cloud Microservices. Combines industrial production engineering experience at LTIMindtree with rigorous computational research, an IEEE publication in pervasive computing, and advanced quantitative training through graduate studies in Financial Engineering at WorldQuant University. Possesses a deep dual-competency in deterministic software systems (Java 17+, Spring Boot, SQL, FastAPI) and probabilistic intelligence (Generative AI, RAG, Reinforcement Learning, Computer Vision), certified across Microsoft, Anthropic, and AWS ecosystems.

MASTER TECHNICAL & FUNCTIONAL COMPETENCIES

Artificial Intelligence & Agentic Systems: Hierarchical Multi-Agent Systems (HMAS), Autonomous AI Agents, Multi-Agent Orchestration, Microsoft Copilot Studio, Azure AI Studio, Anthropic Claude API, Google Gemini API, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Embeddings & Similarity Search, Reinforcement Learning (PPO, SAC), Machine Learning Interpretability, Responsible AI.

Enterprise Automation & Microsoft Ecosystem: Microsoft Power Apps (Canvas & Model-Driven), Power Automate (Complex Multi-Stage Cloud Flows, Error-Handling & Custom Connectors), Microsoft Dataverse (Schema Design & Entity Relations), SharePoint Integration, Power BI (Business Intelligence & Executive Dashboards), Application Lifecycle Management (ALM).

Data Engineering, Analytics & Quantitative Modeling: Exploratory Data Analysis (EDA), Statistical Modeling, ETL Ingestion Pipelines, Feature Engineering, Outlier Analysis & Data Preprocessing, Data Modeling, Pandas, NumPy, Scikit-learn, Relational Query Tuning & Database Indexing (PostgreSQL, MySQL, SQL Server), MongoDB (NoSQL), Quantitative Finance Modeling.

Programming Languages, Frameworks & Backend Systems: Python, Java (17+), FastAPI, Flask, Spring Boot, Spring Security (JWT, Role-Based Access Control / RBAC), WebSockets (60 Hz Low-Latency Streaming), RESTful Microservices, HTML5, CSS3, ReactJS, Tailwind CSS, Data Structures & Algorithms (LeetCode Problem Solving).

Cloud Infrastructure, DevOps & Engineering Practices: Azure Cloud Infrastructure, AWS Cloud Infrastructure, Git, GitHub Actions, CI/CD Automated Pipelines, Telemetry & Performance Tuning, Continuous Integration, Test-Driven Development (TDD), Agile / Scrum Methodology.

PROFESSIONAL EXPERIENCE

LTIMindtree | *Bengaluru, India* (August 2025 – Present)

Role: Software Engineer | AI Solutions, Power Platform & Backend Developer

- **Autonomous Agent Engineering (Copilot Studio):** Architected, configured, and deployed autonomous enterprise AI agents using Microsoft Copilot Studio and generative AI foundation models. Implemented multi-turn conversational orchestration, intent recognition, deterministic tool routing, and retrieval-augmented generation (RAG) across enterprise repositories.
- **Extensible Agent Actions & Azure AI Integration:** Engineered extensible Copilot agent actions and custom plugins utilizing Azure AI services and REST connectors, enabling autonomous multi-step execution for enterprise business workflows and customer self-service.
- **Enterprise Process Automation (Power Platform):** Spearheaded end-to-end digital transformation solutions across Power Apps (Canvas & Model-Driven), Power Automate, and Dataverse, automating complex cross-departmental operations and slashing manual processing overhead by 50%.
- **Production Cloud Flows & Resiliency:** Authored multi-stage automated cloud flows in Power Automate featuring custom connectors, secure API integrations, and robust exception-handling logic, systematically saving ~20 manual operational hours per week.
- **Data Architecture & Optimization:** Integrated Power Platform solutions with underlying relational databases and custom APIs, writing optimized SQL queries and tuning relational schemas that improved enterprise data retrieval times by 40%.
- **Scalable Backend Microservices & Security:** Architected enterprise backend microservices and secure data persistence layers using Java 17+, Spring Boot, and RESTful APIs, maintaining 99.9% uptime. Enforced enterprise security boundaries via JWT authentication and Spring Security Role-Based Access Control (RBAC).

InTrain Tech (October 2023 – November 2023)

Role: Machine Learning & Data Science Intern

- **Credit Risk Scoring Models:** Engineered Python-based predictive machine learning models for automated loan credit score validation, reducing loan risk prediction classification error rates by 15%.
- **Feature Engineering & Hyperparameter Optimization:** Improved overall model evaluation performance by 20% through structured feature selection, outlier handling, and systematic grid/Bayesian hyperparameter optimization, cutting training computation time by 10%.

- **Comparative Algorithm Benchmarking:** Conducted comprehensive comparative analyses across ensemble methods (Random Forest, Gradient Boosting, XGBoost), selecting optimal trade-offs between model interpretability and predictive accuracy.

Ethnus (August 2023 – November 2023)

Role: Full Stack Development Intern (MERN)

- **Secure Web Application Engineering:** Engineered and maintained a full-stack MERN (MongoDB, Express, React, Node.js) web application designed for high-integrity online voting platforms.
- **Security & Access Control:** Implemented stateless JWT-based authentication and role-based route guards, resulting in a 25% reduction in unauthorized session access incidents.

MASTER TECHNICAL PROJECTS PORTFOLIO

AutoFoos: Hierarchical Multi-Agent System (HMAS) | December 2025 – Present

Tech Stack: FastAPI, Google Gemini API, WebSockets, Python, HTML5 Canvas, Applied Physics [Live Application | GitHub Repository]

- Architected an end-to-end web-based Hierarchical Multi-Agent System (HMAS) simulating real-time competitive foosball across 10 autonomous physics agents in a browser environment.
- Integrated supervisory LLM coaches powered by Google Gemini to analyze opponent shot spatial clusters and dynamically issue strategic macro-level play directives.
- Engineered reactive low-level rod agents using real-time raycasting for continuous ball trajectory interception, passing corridors, and defensive anchoring.
- Built a headless 60 Hz physics engine in FastAPI incorporating Magnus spin dynamics, stamina decay, and continuous collision detection, streaming simulation state live via WebSockets.

Digital Twin Technology for Type 2 Diabetes Risk Prediction | December 2024 – April 2025

Tech Stack: Reinforcement Learning (PPO, SAC), XGBoost, Python, Pandas, Scikit-learn

- Developed an AI-driven digital twin framework to simulate individual physiological profiles and predict Type 2 diabetes risk trajectories with 94.5% predictive accuracy.
- Integrated policy-gradient reinforcement learning agents (Proximal Policy Optimization & Soft Actor-Critic) and XGBoost to deliver personalized, adaptive intervention recommendations based on continuous biometric and lifestyle data.

NeuroVision: AI-Driven Brain Tumor Detection & Segmentation | August 2024 – November 2024

Tech Stack: PyTorch, CNN, VGG16, DenseNet, U-Net, Flask, React, Tailwind CSS

- Engineered an automated medical imaging diagnostic pipeline utilizing CNN, VGG16, DenseNet, and U-Net segmentation architectures on brain MRI datasets, achieving 97.14% classification accuracy.
- Deployed the complete inference pipeline via a lightweight Flask backend and React frontend, cutting end-to-end diagnostic inference latency to under 2 seconds.

AI-Based Rice Plant Disease Detection Using Deep Learning | August 2024 – November 2024

Tech Stack: MobileNet, Computer Vision, Flask, React, Tailwind CSS, Python

- Designed a computer vision system to detect rice plant leaf diseases using MobileNet architecture, reaching 99.41% classification accuracy.
- Constructed a lightweight Flask backend and React frontend optimized for low-latency inference in bandwidth-constrained rural agricultural deployments.

Loan Credit Score Predictor & Risk Engine | November 2023 – December 2023

Tech Stack: Random Forest, Python, Scikit-learn, Feature Selection, Data Preprocessing [GitHub Repository]

- Created an ensemble machine learning model to evaluate loan default risk, enhancing prediction accuracy by 15% through correlation analysis, recursive feature elimination, and imputation.
- Optimized training pipelines to cut computation overhead by 12% while ensuring robust out-of-sample generalization.

PEER-REVIEWED PUBLICATIONS & RESEARCH

- **Tankala Sai Prasanth (Co-Author) et al.,** *"AI-Enhanced Smartwatches: Bridging Technology, Behavior, and Continuous Usage"*, IEEE ASSIC, September 2025. Available on IEEE Xplore (DOI: 10.1109/ASSIC.2025.11158655)

Scope & Contribution: Formulated research methodology, led empirical model experimentation and behavioral telemetry analysis, and co-authored manuscript examining continuous usage factors in wearable artificial intelligence.

EDUCATION & ACADEMIC BACKGROUND

Master of Science in Financial Engineering (MSFE) | WorldQuant University (Enrolled / Ongoing | Expected 2027)

Coursework: Stochastic Processes, Quantitative Portfolio Theory, Machine Learning in Finance, Computational Econometrics, Financial Modeling & Risk Engineering.

Bachelor of Technology (B.Tech) in Computer Science and Business Systems | VIT-AP University, Amaravathi, AP (12/2021 – 04/2025)

Academic Distinction: CGPA **8.95 / 10.0**. Core Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Artificial Intelligence, Operating Systems, Cloud Computing, Object-Oriented Programming (Java), Probability & Statistics.

Board of Intermediate Education (MPC - Mathematics, Physics, Chemistry) | Pandit Junior College, Visakhapatnam, AP
(05/2019 – 03/2021)

High Academic Distinction: Score **962 / 1000 (96.2%)**

PROFESSIONAL CERTIFICATIONS

- **Microsoft Certified: Azure AI Apps and Agents Developer Associate** — Microsoft (Issued Oct 2026 · Expires Oct 2027)
- **Claude Certified Developer** — Anthropic (Issued Aug 2026 · Expires Aug 2027)
- **Microsoft Certified: Agentic AI Business Solutions Architect** — Microsoft (Issued August 2026)
- **Microsoft Certified: AI Agent Builder Associate** — Microsoft (Issued August 2026)
- **AWS Certified: AI Practitioner** — Amazon Web Services (Issued June 2026)
- **AWS Certified: Cloud Practitioner** — Amazon Web Services (Issued August 2026)
- **Complete Data Analyst Training** — Udemy (Issued August 2024) — Data Manipulation, Preprocessing, EDA, Visualization

LEADERSHIP & INSTITUTIONAL ENGAGEMENTS

- **Research Team Member | Blockchain Club, VIT-AP University (12/2023 – 12/2024):** Researched decentralized consensus algorithms and emerging smart contract protocols; co-authored 2 technical whitepapers and presented findings during technical symposia to an audience of 100+ students and faculty.
- **Design Team Lead | SEED Club, VIT-AP University (09/2022 – 09/2023):** Directed an 8-member multidisciplinary design team creating promotional media, technical graphics, and visual content for university-wide workshops, tech summits, and club initiatives.